

Prashanthi Balaga

Data Engineer

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PROFESSIONAL SUMMARY:

- Around 6 years of experience in Designing, Developing, and integrating applications using big data applications by using Hadoop ecosystem, Amazon Web Services (AWS) , Microsoft Azure and Google cloud applications.
- Developed and maintained complex Python applications for data analysis and processing.
- Utilized Python for web scraping, data mining, and automation tasks.
- Built scalable web applications using Django and Flask frameworks.
- Integrated RESTful APIs to enhance application functionality.
- Conducted data analysis and created visual reports using Pandas, NumPy, and Matplotlib.
- Developed dashboards and data visualization tools using Plotly and Dash.
- Automated routine tasks and workflows using Python scripts, improving efficiency by 95%.
- Designed and implemented custom scripts for data cleaning and preprocessing.
- High-level skilled engineer with strong analytical experience, problem solving and understanding business requirements.
- Hands-on experience in installation, configuration and developing the big data infrastructure using the Hadoop clusters on both AWS and Azure cloud platforms.
- Experience in building centralized data warehouse on AWS platform using MySQL database on RDS and S3.
- Good working knowledge on Snowflake and Teradata databases.
- Expertise in Python data extraction and data manipulation, and widely used python libraries like NumPy, Pandas, and Matplotlib for data analysis.
- Implementing data movement from File system to Azure Blob storage using python API.
- Experience on working with file structures such as text, CSV, JSON, sequence, parquet, and Avro file formats.
- Have Experience in List Comprehensions and Python inbuilt functions such as Map, Filter, and Lambda.
- Expert with BI tools like Tableau and PowerBI, data interpretation, modeling, data analysis, and reporting with the ability to assist in directing planning based on insights.

TECHNICAL SKILLS:

Programming Languages:	Python, PySpark, Spark, Scala, Shell scripting, Java
Cloud Technologies:	AWS, GCP, Azure, Snowflake
IDEs/Development Tools:	PyCharm, VScode, PyStudio, Eclipse, IntelliJ
Databases:	MySQL, Aws RDS, Snowflake, Teradata, SQL Server, Oracle, Postgre SQL
No SQL Databases:	Mongo DB, Hadoop HBase, Apache Cassandra
Operating System:	Windows, Linux, UNIX
Querying Languages:	SQL, NoSQL, My SQL, Spark SQL
Big Data Eco System:	Spark, SparkSQL, Spark Streaming, PySpark
Visualization & Reporting Tools:	Tableau, Power BI, Matplotlib, Seaborn

PROFESSIONAL EXPERIENCE:

Client: AbbVie, IL

Aug 2023 – Present

Role: Data Engineer

Responsibilities:

- Worked on AWS cloud platform and its features, which include Textract service, EMR, EC2, S3, VPC, SNS,RDS and Cloud Watch.
- Implemented a CI/ CD pipeline with Docker, and GitHub by virtualizing the servers using Docker for the Dev and Test environments by achieving needs through configuring automation using containerization.
- Utilize Python, PostgreSQL, GitHub, and AWS Cloud services for project development.
- Experienced Python Developer with over 5 years of expertise in building scalable web applications and automating complex processes. Adept at writing clean, efficient code and collaborating with cross-functional teams to deliver high-quality software solutions.

- Use AWS Textract to convert PDF content and to present the data using UI framework.
- Code using PyCharm to transform files from PDF to CSV and then load the data into an application.
- Transfer data and metadata from Excel to tables for viewing.
- Build a basic ETL process to ingest record data from a web application with thousands of active users daily.
- Write ETL scripts using PySpark in AWS Glue, leveraging its serverless and cost-effective nature.
- Use Amazon RedShift to perform SQL analytics on the data.
- Ensure efficient object storage and data processing using AWS S3 Bucket and S3DistCp for copying data to HDFS.
- Employ additional AWS services such as EC2, IAM, and Data Pipeline for seamless operations.
- Manage code repositories using GitHub to ensure version control and collaborative development.
- Deploy and manage applications using AWS EC2 instances to facilitate code execution and storage.
- Implement data validation and cleaning processes to ensure accuracy and reliability of extracted data.
- Conduct regular audits to maintain data integrity throughout the ETL lifecycle.
- Optimize ETL processes to enhance performance and reduce latency in data processing.
- Implement best practices for cost management and resource allocation within AWS services.
- Ensure compliance with data protection standards and policies by managing access through AWS IAM.
- Implement security measures to protect sensitive information handled in the COA project and associated data services.
- Created Tableau Score cards, Dashboards using stack bars, bar graphs, scatter plots, geographical maps, and Gantt charts.
- Conducted in-depth data analysis using Tableau to identify trends, patterns, and outliers in large datasets.
- Translated business requirements into technical specifications to develop meaningful data visualizations.
- Integrated Tableau with various data sources such as SQL databases, cloud services, and Excel to ensure seamless data flow.
- Write and maintain comprehensive project documentation.
- Collaborate with the business team to verify requirements and ensure alignment with project objectives.
- Create training content for new hires to efficiently onboard and ramp up their participation in the project.
- Foster team collaboration to enable effective learning and skill enhancement to achieve project goals.
- Communicate effectively with cross-functional teams including data scientists, analysts, and IT support.
- Participate in project meetings and provide updates on progress, challenges, and solutions.
- Identify opportunities for process automation and improvement to enhance efficiency.
- Keep abreast of the latest trends in cloud computing and data processing to introduce innovative techniques into the project.

Client: US Bank, MO

July 2021 – July 2023

Role: AWS Data Engineer

Responsibilities:

- Worked on AWS cloud platform and its features, which includes EMR, Glue, EC2, S3, VPC, SNS, RDS and Cloud Watch.
- Implemented a CI/ CD pipeline with Docker, Jenkins, and GitHub by virtualizing the servers using Docker for the Dev and Test environments by achieving needs through configuring automation using containerization.
- Configured and deployed highly available and scalable applications on AWS cloud using EC2, RDS, S3, ELB, Autoscaling and monitored these applications with Cloud watch, Grafana and AWS SNS.
- Built and deployed docker containers to break up monolithic apps into microservices, improving developer workflow, increasing scalability, and optimizing speed using Kubernetes clusters in AWS Elastic Container Service.
- Got involved in migrating on prem Hadoop system to using GCP (Google Cloud Platform).
- Migrated previously written cron jobs to airflow/composer in GCP.
- Developed data loading strategies using the Hadoop cluster (HDFS, HIVE, AWS kinesis)
- Created GCP Big Query authorized views for row level security or exposing the data to other teams.
- Have Extensive Experience in IT data analytics projects, Hands on experience in migrating on premise ETLs to Google Cloud Platform (GCP) using cloud native tools such as BIG query, Cloud Data Proc, Google Cloud Storage, Composer.
- Built and deployed Docker containers to break up monolithic applications into micro services, improving developer workflow, increasing scalability, and optimizing speed using Kubernetes clusters in AWS Elastic Container Service (ECS).

- Deployed Cloud IAM solution for auditing purposes such as to find out what level of access each of the AWS users in the organization has across all the projects.
- Created data sets for developing the azure data factory pipelines and maintained the architectural responsibilities.
- Generated, wrote, and executed SQL scripts in big query to make changes including updates, additions, creations, and bank procedures.
- Worked on Big Data Hadoop cluster implementation and data integration in developing large-scale system software.
- Worked on optimizing infrastructure operations of Google Cloud Platform by designing AWS infrastructure, leveraging the latest services and innovations from AWS.
- Work closely with AWS Platform Engineer to incorporate solutions into custom designed cloud-based applications.
- Designed and managed Azure data factory pipelines and pulled data from SQL server, Google cloud.
- Developed data warehouse using Star schema/Snowflake schema concepts in Snowflake using Snow SQL.
- Worked on Cloud SQL, Cloud Composer AWS Big query, and source Data base.
- Created with Hue UI in scheduling jobs with ease and File browsing, Job browsing, Megastore management.
- Worked extensively with Sqoop for importing and exporting the data from HDFS to Relational Database systems / mainframe and vice-versa loading data into HDFS.
- Used Apache airflow in AWS composer environment to build data pipelines and used various airflow operators like bash operator, Hadoop operators and python callable and branching operators.

Client: AT&T, TX

Jan 2019 – June 2021

Role: Data Analyst

Responsibilities:

- Experience in working with structured data by importing and exporting from DynamoDB to HDFS, Hive using Sqoop.
- Involved in data migration from existing data platforms to Hadoop and built data warehouse within Hadoop clusters such as Hive, oozie and Sqoop.
- Used flume to generate data cluster files and loaded the data to Relational database management systems by using Sqoop.
- Involved in migration process of Teradata from SQL server to Snowflake.
- Designed and implemented data pipeline framework to for data ingestion to snowflake.
- Experience in creating database objects in Snowflake.
- Write ETL (Extract / Transform / Load) processes, design database systems and develop tools for real-time and offline analytic processing.
- Designed ETL processes to load prospects to data marts.
- Worked on the ETL process using SQL to populates data from the database servers.
- Created and maintained Logical Data Model (LDM) for the project. Includes documentation of all entities, attributes, data relationships, primary and foreign key structures, allowed values, codes, business rules, glossary terms, etc.
- Worked on cloudera distribution and integrated Hadoop clusters to cloudera distribution system.
- Developed and maintained MySQL database structures in test, quality assurance, and production environments utilizing current standards and procedures.
- Created Tableau Score cards, Dashboards using stack bars, bar graphs, scatter plots, geographical maps, and Gantt charts.
- Integrated Tableau to existing databases like MySQL to run background jobs.
- Created Tableau views with complex calculations and hierarchies making it possible to analyse and obtain insights into large data sets.

Client: Tredence , IA

April 2019 – Dec 2019

Role: Data Analyst

Responsibilities:

- Experience in admin responsibilities for SQL server for various cluster environments using inbuilt tools (Query store, SQL server profiler).

- Been part of change management processes and created users in the database system as per the requirements.
- Responsible for providing access for different role groups for various departments Gathered the User Requirements from the Business Users and the Business Sponsor.
- Created user logins with integrated the single sign on with the existing IT framework.
- Managed permissions and access for overall organizational hierarchy and allocated privileges based on the managers and teams across the company's retail store employees.
- Assisted the teams to migrate databases by importing, exporting and database mirroring.
- Integrated the SQL server to APX tool to ensure data security compliances were managed.
- Maintained APX servers for generating reports and assigned access for Management heads, sales heads and retail store managers based on their reporting hierarchy.
- Involved in the database recovery and backup process for the organizations and provided support to users to troubleshoot issues.
- Performance analysis of batch jobs by using Spark Tuning parameters.
- Experience in CI/CD tool Jenkins for code deployment and scheduling of jobs.
- Involved in writing the SQL queries using oracle and creating the Materialized views.